NAVAL MEDICAL RESEARCH UNIT SAN ANTONIO
PERFORMANCE WORK STATEMENT

BioAssemblyBot500 Equipment Contract

1. Scope of Work

The Naval Medical Research Unit San Antonio (NAMRU-SA) conducts gap driven combat casualty care, craniofacial, and directed energy research to improve survival, operational readiness, and safety of Department of Defense personnel engaged in routine and expeditionary operations. An equipment contract is required to procure 1-Single Channel Pipette BioAssemblyBot (BAB) Hand, 1-BioAssemblyBot 8-Channel Pipette, 1-BioAssemblyBot 500 8-Channel Liquid Handling Bundle, 1-Single Cell Dispense BioAssemblyBot Hand and on-site services for installation, configuration or training, including all travel costs. The equipment is essential for R&D projects that will be executed by NAMRU-SA in FY 2025. This equipment will enhance and support interoperability with existing equipment (BioAssemblyBot 500 Model: NU-L543-400, Manufacturer: Advanced Solutions Life Sciences) currently in use at NAMRU-SA.

2. Definitions and Terms

The Single Channel Pipette BioAssemblyBot hand must have vertical dispense with automatic detection of liquid levels, tips, blockage, step loss. Compatible with standard pipette tips including clear or black (conductive) with 10, 20, 25, 50, 175, 200, and 1,000 µL capacities. Volume resolution of 0.024µL/step with 48000 increments/full stroke.

The BioAssemblyBot 500 8-Channel Liquid Handling Bundle must have dual bulk reservoirs to store and dispense large volumes of liquid, such as buffers, reagents, or media. They must have a capacity of 50 mL each and can be easily refilled or replaced. Dual tip rack that accommodates up to 96 disposable tips each. Tip stripper that allows the BioAssemblyBot to automatically remove used tips quickly and safely from the 8-Channel Pipette BioAssemblyBot Hand, without touching them. Tip waste collection bin that collects the discarded tips from the tip stripper and has a capacity of 192 tips

The Single Cell Dispense BioAssemblyBot Hand must be able to deliver single cells into wells. It must be BioApps certified, with a ≥ 75% single cell efficiency rate, and ≤ 12 min dispense time for 96-well plates. A BioAssemblyBot standard stage overlay with recessed well plate positioning, cell source station and tip wash station must be provided.

3. Support Requirements

The contracted company shall provide an equipment contract as required by NAMRU-SA. The vendor selected for the contract of this Performance Work Statement must have the ability to provide 1-Single Channel Pipette BioAssemblyBot Hand, 1-BioAssemblyBot 8-Channel Pipette, 1 BAB 500 8-Channel Liquid Handling Bundle, 1-Single Cell Dispense BioAssemblyBot Hand, and on-site services for installation, configuration or training, including all travel costs.

4. Additional Requirements/Qualifications

No additional requirements.

5. Delivery Address

Naval Medical Research Unit San Antonio
4141 Petroleum Road, Bldg. 3260
JBSA Fort Sam Houston, TX 78234

6. Contract Period of Performance

The anticipated period of performance is 12 months.